

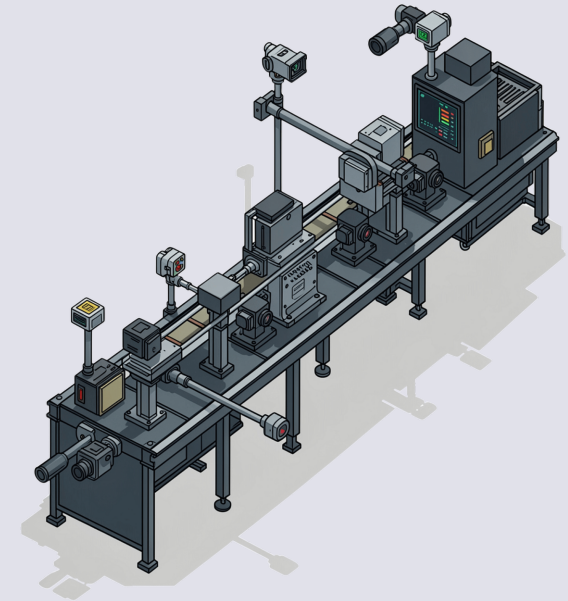
Manufacturing Quality Intelligence System (MQIS)

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The Problem: Manual QC Doesn't Scale

Slow & Inconsistent

Human inspectors check each part visually — unable to keep pace across parallel production lines.

Detection Without Diagnosis

Existing automated systems flag defects but cannot explain *why* they are happening.

No Real-Time Visibility

Supervisors review data end-of-shift, missing live failures as they compound on the floor.

Four Concrete Challenges

Detection ≠ Diagnosis

A defect is flagged, but there is no connection to its root cause or emerging pattern.

No Real-Time Visibility

A failing machine can produce dozens of rejects before anyone on the floor notices.

Knowledge is isolated

Technician expertise lives in heads and paper logs — never captured inside the system.

Manual Reporting

Shift reports are time-consuming and pull skilled technicians away from the production floor.

Our Four-Layer Solution



1 Vision-based Detection

Image differencing and contour analysis classifies defects — bent, color, scratch — with severity and confidence score.

2 Event Logging

Every inspection is logged with timestamp, line ID, defect type, and severity for full traceability.

3 Live Analytics

Real-time per-line rejection rates, status flags (Normal / Warning / Critical), and hourly defect breakdowns.

4 RAG-based Explanation

LLM retrieves curated factory knowledge, combines it with live data, and generates grounded shift reports and root-cause insights.

The Innovation: From Detection to Understanding

1

Detect

Vision system flags defects instantly, in real time, on every production line.

2

Understand

RAG grounds LLM outputs in real data and curated factory knowledge — **no hallucinations.**

3

Act

Supervisors receive explainable answers in plain English, not black-box flags or raw logs.



Future roadmap: extend SentinelLine to predictive maintenance by leveraging early trend detection from accumulated event logs.

